MARKING SCHEME

Model Paper — 06

Mathematics II

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Note (for teachers):

- Candidates answer **five** questions from Part A (Questions 1–6) and **five** questions from Part B (Questions 7–12). Each question carries 10 marks; total 100 marks.
- The marks for each step are given in the right-hand column and add up to the marks printed on the paper for each part.
- Give full marks for correct alternative methods. Accept reasonable readings from graphs and constructions (acceptance ranges are given), and answers that agree with 4-figure table values to the last digit.

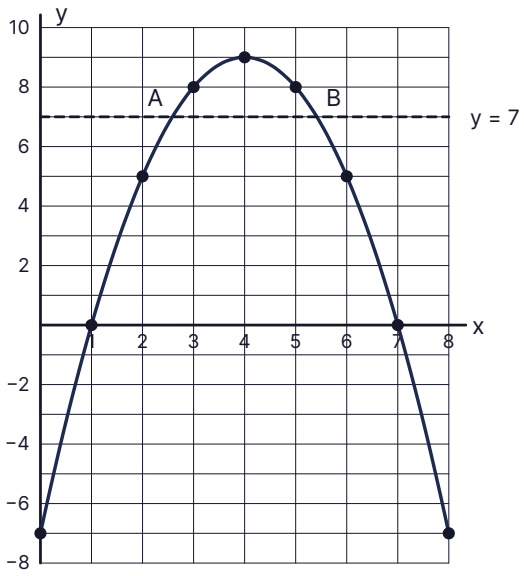
Part A — Answers

Question 1

(i)	
Interest = 60 000 × $\frac{18}{100}$ × $\frac{10}{12}$	**1**
= Rs 9 000	**1**
(ii)	
Total paid = 6 550 × 10 = Rs 65 500	**1**
Total interest = 65 500 − 60 000 = Rs 5 500	**1**
(iii)	
Balance owed at the start of each month: Rs 60 000, 54 000, 48 000, 42 000, 36 000, 30 000, 24 000, 18 000, 12 000, 6 000 (the repaid capital is 60 000 ÷ 10 = Rs 6 000 per month)	**2**
Sum of the monthly balances = 6 000 × (10 + 9 + … + 1) = 6 000 × 55 = Rs 330 000 (rupee-months)	**1**
Let the annual rate be r %: 5 500 = 330 000 × $\frac{r}{100}$ × $\frac{1}{12}$	**1**
r = 20 %	**1**
Option 1 rate = 18 % and 20 % > 18 %, so the annual interest rate under Option 2 is higher (although its interest, Rs 5 500, is less than Rs 9 000)	**1**

Total marks: 10

Question 2



(i)	
The axis of symmetry is $x = 4$ ($y = 5$ at $x = 2$ and $x = 6$); the point $x = 5$ matches $x = 3$, so $y = 8$	**1**
(ii)	
Suitable scale marked on both axes (2 cm per unit on x; 2 cm per 2 units on y)	**1**
All nine points (0, −7), (1, 0), (2, 5), (3, 8), (4, 9), (5, 8), (6, 5), (7, 0), (8, −7) plotted (8 or 9 correct: 1 mark; fewer: 0)	**1**
Smooth curve (an inverted U-shape) through the points, with maximum at (4, 9), see figure	**1**
(iii)	
(a) $x = 1$ and $x = 7$	**1**
(b) $y = -(x - 1)(x - 7)$ ($p = -1$, $q = -7$)	**1**
(iv)	
Line $y = 7$ drawn; it meets the curve at A (≈ 2.6, 7) and B (≈ 5.4, 7)	**1**
(a) $2.6 < x < 5.4$ (accept 2.5–2.7 and 5.3–5.5)	**1**
(b) AB = 5.4 − 2.6	**1**
= 2.8 units (accept 2.7 – 2.9; exact value $2\sqrt{2} \approx 2.83$)	**1**

Total marks: 10

Question 3

(i)	
In triangle ABC ($\angle B = 90°$): BC = AC sin 35° = 40 × 0.5736	**1**
= 22.94 m	**1**
(ii)	
BD = 2 × BC = 2 × 22.94 = 45.88 m (since DC = BC)	**1**
(iii)	
$\tan \angle DEB = \frac{BD}{BE}$	**1**
$= \frac{45.88}{36} = 1.2744$	**1**
From the tangent table: tan 51°48′ = 1.2708 and tan 51°54′ = 1.2753; 1.2744 lies between them (difference 0.0036 from 1.2708, i.e. about 5′)	**1**
$\angle DEB$ = 51°53′ (accept 51°52′ – 51°54′)	**1**
(iv)	
Inclination of DE = $\angle DEB$ = 51°53′ ; inclination of AC = $\angle CAB$ = 35°	**1**
Difference = 51°53′ − 35° = 16°53′	**1**
16°53′ > 16°, so the inclination of DE exceeds that of AC by more than 16°	**1**

Total marks: 10

Question 4

(i)	
Area of the rectangle = $(2x + 2)(x + 1) = 2x^2 + 4x + 2$; area of the pond = $(x - 1)^2 = x^2 - 2x + 1$	**1**
$2x^2 + 4x + 2 - (x^2 - 2x + 1) = 100 \Rightarrow x^2 + 6x + 1 = 100 \Rightarrow x^2 + 6x - 99 = 0$	**1**
$x = \frac{-6 \pm \sqrt{(36 + 396)}}{2} = \frac{-6 \pm \sqrt{432}}{2}$ (formula with $a = 1$, $b = 6$, $c = -99$)	**1**
$\sqrt{432} = 12\sqrt{3}$, so $x = -3 \pm 6\sqrt{3}$; reject the negative root	**1**
$x = -3 + 6 \times 1.73 = 7.38 \approx 7$ m	**1**
(ii)	
Side of the pond = $x - 1$ = 6 m	**1**
Diagonal = $\sqrt{(6^2 + 6^2)} = 6\sqrt{2}$	**1**
= 6 × 1.41 = 8.46 m	**2**
8.46 > 8, so the diagonal is longer than 8 m	**1**

Total marks: 10

Question 5

(i)	
$x + y = 80$ .....①	**1**
$95x + 150y = 9\,250$ ($\div 5$: $19x + 30y = 1\,850$) .....②	**1**
From ①, $x = 80 - y$; substituting in ②: $19(80 - y) + 30y = 1\,850$	**1**
$1\,520 + 11y = 1\,850 \Rightarrow 11y = 330 \Rightarrow y = 30$	**1**
$x = 50$: 50 small packs and 30 large packs	**1**
(ii)	
24.5 kg = 24 500 g	**1**
Let a small pack contain m g; a large pack contains $(m + 150)$ g	**1**
$50m + 30(m + 150) = 24\,500$	**1**
$80m + 4\,500 = 24\,500 \Rightarrow 80m = 20\,000$	**1**
$m = 250$ g	**1**

Total marks: 10

Question 6

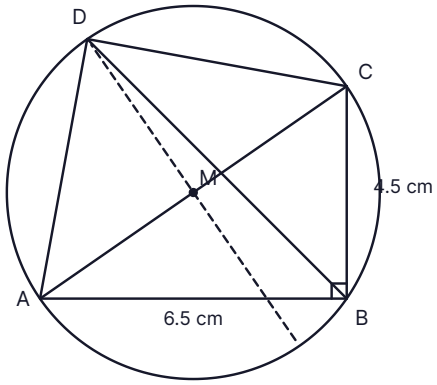
Class	Mid-value x	Frequency f	fx
15–24	19.5	4	78
25–34	29.5	9	265.5
35–44	39.5	14	553
45–54	49.5	9	445.5
55–64	59.5	4	238
		40	1 580

(i)	
Mid-values 19.5, 29.5, 39.5, 49.5, 59.5 and products fx (see table)	**1**
$\Sigma f = 40$ and $\Sigma fx = 1\,580$	**1**
Mean = $\frac{1\,580}{40}$ = 39.5 patients per day	**1**
(ii)	
Expected number = 39.5 × 30	**1**
= 1 185 patients	**1**
(iii)	
Expected patients in 20 days = 39.5 × 20 = 790	**1**
Total fees = 790 × 250 = Rs 197 500	**1**
Hospital's share = $\frac{40}{100}$ × 197 500 = Rs 79 000	**1**
(iv)	
Fewer than 35: classes 15–24 (4 days) and 25–34 (9 days); minimum when each day has the smallest value in its class (15 and 25)	**1**
Minimum total = 4 × 15 + 9 × 25 = 60 + 225 = 285 patients	**1**

Total marks: 10

Part B — Answers

Question 7



(i)	
AB = 6.5 cm drawn accurately (± 1 mm)	**1**
Perpendicular at B constructed with compass (arcs visible), ∠ABC = 90°	**1**
C marked with BC = 4.5 cm (± 1 mm); AC joined (AC ≈ 7.9 cm)	**1**
(ii)	
Arcs of equal radius from A and from C on both sides of AC	**1**
Perpendicular bisector drawn; it meets AC at its midpoint M (≈ 3.95 cm from A)	**1**
Circle with centre M and radius MA (≈ 3.95 cm) drawn; it passes through A, B and C	**1**
(iii)	
Bisector of ∠ABC constructed (45° each side, arcs visible)	**1**
D marked on the circle (BD meets the circle again); AD and CD joined. AD = CD ≈ 5.6 cm (5.5 – 5.7 accepted)	**1**
(iv)	
∠ADC = 90° (angle in a semicircle, AC is a diameter)	**1**
∠DAC = ∠DBC = 45° (angles in the same segment; BD bisects the right angle ABC)	**1**

Total marks: 10

Question 8

(i)	
In triangles BGE and CFE: $\angle GBE = \angle FCE$ (alternate angles, AG ∥ DC)	**1**
BE = EC (E is the midpoint of BC)	**1**
$\angle BEG = \angle CEF$ (vertically opposite angles); therefore $\triangle BGE \cong \triangle CFE$ (angle, side, angle)	**1**
(ii)	
From the congruence, GE = EF and BE = EC, so the diagonals BC and GF of the quadrilateral BGCF bisect each other at E	**1**
Therefore BGCF is a parallelogram. (Alternatively: BG = CF and BG ∥ CF.)	**1**
(iii)	
In triangles DFH and BGE: $\angle HDF = \angle HAB$ (corresponding angles, DC ∥ AB) $= \angle EBG$ (corresponding angles, AH ∥ BC)	**1**
$\angle DFH = \angle BGE$ (alternate angles, DC ∥ AG); the third angles are equal too, so the triangles are equiangular	**1**
(iv)	
Similar triangles: $\frac{DH}{BE} = \frac{DF}{BG}$	**1**
$BG = CF = DC - DF = \frac{5}{6}\,DC$; $DF = \frac{1}{6}\,DC$; $BE = \frac{1}{2}\,BC = \frac{1}{2}\,AD$	**1**
$DH = \frac{1}{2}\,AD \times \frac{(1/6)\,DC}{(5/6)\,DC} = \frac{1}{2}\,AD \times \frac{1}{5} = \frac{1}{10}\,AD$	**1**

Total marks: 10

Question 9

(i)	
Volume of a pyramid = $\frac{1}{3} \times (3a)^2 \times 2a = 6a^3$	**1**
Volume of 4 pyramids = $24a^3$	**1**
Volume of the cylinder = $\pi r^2 \times 8r = 8\pi r^3$	**1**
$24a^3 = 8\pi r^3 \Rightarrow a^3 = \frac{\pi}{3} r^3$	**1**
(ii)	
$\lg a^3 = \lg 3.14 + 3 \lg 14.5 - \lg 3 = 0.4969 + 3 \times 1.1614 - 0.4771$	**1**
$= 0.4969 + 3.4842 - 0.4771 = 3.5040 \Rightarrow a^3 = 3\,191$ (accept 3 190 – 3 192)	**1**
$\lg a = 3.5040 \div 3 = 1.1680 \Rightarrow a = 14.72 \approx 15$ cm	**1**
(iii)	
Volume of the cuboid = $3a \times 3a \times 2a = 18a^3$; volume cut out = $6a^3$	**1**
Remaining volume = $12a^3$	**1**
$= 12 \times 3\,191 = 38\,292$ cm^3 (accept 38 280 – 38 304)	**1**

Total marks: 10

Question 10

(i)	
120, 130, 140 (m^2)	**1**
(ii)	
$T_{12} = a + 11d = 120 + 11 \times 10$	**1**
$= 230\ m^2$	**1**
(iii)	
$S_{12} = \frac{12}{2}(120 + 230)$	**1**
$= 2\ 100\ m^2$	**1**
(iv)	
$S_{14} = \frac{14}{2}(2a + 13 \times 20) = 2\ 100$	**1**
$2a + 260 = 300$	**1**
$a = 20\ m^2$	**1**
(v)	
Day n: $120 + (n - 1) \times 10 = 20 + (n - 1) \times 20$	**1**
$110 + 10n = 20n \Rightarrow n = 11$: the 11th day (both transplant 220 m^2)	**1**

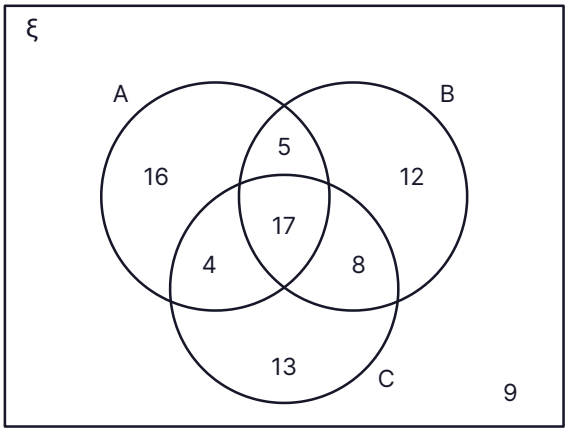
Total marks: 10

Question 11

(i)	
TA ⊥ OA and TB ⊥ OB (a tangent is perpendicular to the radius at the point of contact): ∠TAO = ∠TBO = 90°	**1**
∠TAO + ∠TBO = 180° (opposite angles of TAOB)	**1**
Therefore TAOB is a cyclic quadrilateral (opposite angles supplementary)	**1**
In triangles TAO and TBO: ∠TAO = ∠TBO = 90°, OA = OB (radii), OT common (RHS)	**1**
△TAO ≅ △TBO, so ∠AOT = ∠BOT: OT bisects ∠AOB (also TA = TB)	**1**
(ii)	
(a) ∠TBA = ∠TAB, because TA = TB (tangents from T; shown by the congruence in (i)), so triangle TAB is isosceles	**1**
(b) ∠ADB = ∠TAB (angle in the alternate segment: tangent TA, chord AB)	**2**
(c) ∠AOB = 2∠ADB (angle at the centre is twice the angle at the circumference)	**1**
OT bisects ∠AOB, so ∠AOT = $\frac{1}{2}$∠AOB = ∠ADB = ∠TAB	**1**

Total marks: 10

Question 12



(i)	
Farmers who grew at least one crop = 84 − 9 = 75	**1**
Two or more crops: 34	**1**
Exactly one crop = 75 − 34 = 41	**1**
(ii)	
Let the number who grew all three = t; exactly two = t	**1**
$t + t = 34$	**1**
$t = 17$	**1**
(iii)	
Exactly two = 17: 5 (paddy and maize only) + $p + 2p = 17 \Rightarrow 3p = 12 \Rightarrow p = 4$	**1**
Paddy and vegetables only = 4 ; maize and vegetables only = 8	**1**
(iv)	
n(A) = only paddy + 5 + 4 + 17 = 42	**1**
Only paddy = 42 − 26 = 16 (only maize + only vegetables = 41 − 16 = 25; the data do not separate them, so the figure shows 12 and 13 for illustration)	**1**

Total marks: 10